Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

DISASTER MANAGEMENT

The domain of the Project:

FULL STACK DEVELOPMENT

Project Done By:

U Ashok Kumar - G30 FSD

Period of the project

March 2026 to April 2026

(12/03/2026 - 30/04/2026)

Declaration

The project titled "Disaster Management" has been mentored by Ms.Akshara(Software Developer at Propel Technology), organised by SURE Trust, from March 2026 to April 2026, for the benefit of the educated unemployed rural youth for gaining hands-on experience in working on industry relevant projects that would take them closer to the prospective employer. I declare that to the best of my knowledge that I have worked on it successfully and enhanced the practical knowledge in the domain.

Member:

Mr. U Ashok Kumar

Mentor's Name
Ms.Akshara
(Software Developer at Propel Technology)

Prof. Radhakumari
Executive Director & Founder
SURE Trust

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Executive Summary

The **Disaster Management System** is a comprehensive web-based platform developed to strengthen disaster preparedness, emergency response, resource coordination, and post-disaster recovery activities. Natural and man-made disasters such as floods, cyclones, earthquakes, fires, landslides, industrial accidents, and pandemics continue to create major threats to human life, public infrastructure, and economic stability. In many cases, the impact of disasters becomes more severe due to delayed communication, lack of organized response systems, poor coordination between agencies, and limited access to reliable information.

Traditional disaster management processes often depend on phone calls, manual records, isolated control rooms, and fragmented communication channels. These approaches can result in slow response times, duplication of efforts, poor resource utilization, and confusion among the public during emergencies. Therefore, there is a growing need for a centralized digital solution that can improve communication, coordination, monitoring, and decision-making during crisis situations.

The proposed Disaster Management System addresses these challenges by offering a modern web-based platform where citizens can report incidents, request emergency help, receive alerts, and track rescue progress. Simultaneously, government departments, emergency teams, hospitals, and administrators can monitor incidents, assign rescue units, manage shelters, track resources, and generate reports through a unified dashboard.

The system is developed using Angular for the frontend interface, Django for backend processing and REST APIs, and PostgreSQL for structured and secure data storage. Cloud deployment can be achieved through Vercel for frontend hosting and Render for backend hosting.

The major goal of the project is to reduce disaster response time, improve transparency, optimize resource usage, and increase public safety. This project demonstrates the practical application of full-stack web technologies in solving critical real-world emergency management challenges.

Background and Context of the Project

Disasters are sudden events that disrupt communities and cause widespread destruction to life, property, environment, and infrastructure. India and many other countries frequently experience floods, cyclones, droughts, fires, road accidents, industrial hazards, earthquakes, and disease outbreaks. With increasing urbanization, climate change, and population growth, the frequency and intensity of disasters are rising significantly.

During emergency situations, timely communication and rapid response are the most important factors in saving lives and reducing losses. However, traditional disaster management methods are often outdated and inefficient. Information may pass slowly through multiple departments, records may be maintained manually, and citizens may not know where to seek help or receive updates.

Digital transformation offers an opportunity to improve disaster management through centralized online platforms. By integrating reporting systems, resource management tools, dashboards, alerts, and communication channels, authorities can respond faster and more effectively.

The Disaster Management System is designed to meet this need by creating a user-friendly, scalable, and secure web application that supports disaster reporting, emergency coordination, and relief management.

Problem Statement

Existing disaster management systems face several operational challenges:

- Delay in receiving emergency reports from affected citizens
- Lack of centralized communication between agencies
- Poor coordination of rescue teams and volunteers
- Difficulty in locating shelters and relief centers
- Manual allocation of emergency resources
- No transparent tracking of rescue requests
- Limited public awareness during emergencies
- Difficulty in generating reports and analytics

These issues can lead to increased casualties, delayed assistance, and inefficient disaster recovery efforts.

Goals of the Project

The primary goals of the Disaster Management System are:

- To provide a digital platform for instant disaster reporting
- To enable real-time communication between citizens and authorities
- To manage rescue teams, shelters, and emergency resources efficiently
- To provide alerts, announcements, and updates during emergencies
- To improve transparency and tracking of rescue operations
- To reduce disaster response time and losses
- To create a scalable system for future smart integrations

The Disaster Management System is designed with clearly defined objectives to improve disaster preparedness, emergency response, communication, and recovery operations through the use of modern web technologies. The project aims to create a reliable, efficient, and scalable platform that helps citizens, rescue teams, and government authorities coordinate effectively during crisis situations.

The objectives of this project are focused on solving the major problems faced in traditional disaster management systems such as delayed reporting, poor coordination, lack of transparency, and inefficient resource utilization.

Clearly Defined Objectives and Goals of the Project

1. To Develop a Centralized Disaster Management Platform

The primary objective of the project is to create a single digital platform where all disaster-related activities such as incident reporting, rescue requests, shelter management, and emergency coordination can be handled efficiently.

2. To Enable Real-Time Incident Reporting

The system allows citizens to report disasters such as floods, fires, earthquakes, cyclones, road accidents, or other emergencies instantly through an online interface. This helps authorities receive information quickly and respond faster.

3. To Improve Communication Between Citizens and Authorities

One of the major goals is to reduce communication gaps during emergencies. The platform enables direct interaction between affected people, rescue teams, administrators, and government departments.

4. To Provide Emergency Alerts and Notifications

The project aims to provide timely alerts, warnings, and announcements to users regarding weather conditions, evacuation notices, rescue progress, or emergency instructions.

5. To Manage Resources Efficiently

The system is designed to track and manage critical emergency resources such as:

- Rescue teams
- Ambulances
- Food supplies
- Medical aid
- Shelter facilities
- Transportation units

Efficient resource management helps save time and reduce losses.

6. To Maintain Shelter and Relief Center Information

The project aims to provide information about available shelters, relief camps, hospitals, and emergency service centers so that affected citizens can quickly find safe locations.

7. To Provide Transparency and Tracking

Users can track the status of their emergency requests or reported incidents. This improves trust, accountability, and transparency in rescue operations.

8. To Build a Scalable and Secure System

The project uses Angular, Django, and PostgreSQL to create a secure and scalable application that can be expanded in the future.

9. To Support Smart Governance and Digital Transformation

The system supports modern governance by replacing manual emergency management methods with an automated digital platform.

📦 Expected Outcomes and Deliverables

After successful development and deployment, the project is expected to produce the following outcomes and deliverables.

🔹 Functional Outcomes

1. Fully Operational Web Application

A complete web-based Disaster Management System that can be accessed through desktop or mobile browsers.

2. User Registration and Login Module

Secure authentication system for citizens, staff members, rescue teams, and administrators.

3. Disaster Reporting Module

Users can submit disaster reports with necessary details such as location, type of incident, severity, and description.

4. Rescue Request Management

Citizens can request urgent assistance, and administrators can assign rescue teams accordingly.

5. Shelter and Relief Management

Information about shelters, hospitals, relief camps, and available facilities will be accessible.

6. Dashboard for Authorities

Administrative dashboard for monitoring incidents, tracking rescue requests, and managing resources.

7. Notification System

Real-time alerts and updates for users during emergencies.

🔹 Technical Deliverables

1. Frontend Application

Responsive and dynamic frontend built using Angular.

2. Backend REST APIs

Secure APIs developed using Django for managing data and system logic.

3. Database Design

Structured database using PostgreSQL to store incidents, users, shelters, and resources.

4. Cloud Deployment

Deployment-ready project hosted using:

- Vercel for frontend

- Render for backend

🔹 Documentation Deliverables

- Project Report
- Source Code Documentation
- Database Schema
- API Documentation
- User Manual
- GitHub Repository Link
- Deployment URL

🔹 Performance Outcomes

The successful implementation of the project is expected to achieve:

- Faster disaster response times
- Better communication and coordination
- Efficient resource utilization
- Increased public safety
- Reduced confusion during emergencies
- Improved disaster preparedness and recovery

The **Disaster Management System** was developed using a structured software development approach to ensure that the final application is reliable, scalable, secure, and user-friendly. The methodology combines planning, system design, frontend development, backend implementation, database integration, testing, and deployment. The final outcome is a web-based platform that enables real-time disaster reporting, emergency coordination, and resource management.

This section explains the methods followed during development, technologies used, software tools utilized, data collection process, project architecture, final results, screenshots explanation, and project repository details.

🔹 Methods / Technology Used

The project was developed using **full-stack web development technologies**, where the frontend, backend, and database are integrated to build a complete and efficient disaster response system.

🔹 Frontend Development

The frontend of the system is developed using Angular. Angular is a modern TypeScript-based framework used for building responsive and dynamic single-page web applications.

Frontend Features:

- Responsive user interface
- Form handling for disaster reporting
- Dashboard views for users and administrators
- Real-time updates display
- Component-based modular structure
- Fast navigation without page reloads

🔹 Backend Development

The backend is developed using Django along with Django REST Framework for creating APIs.

Backend Responsibilities:

- User authentication and authorization
- Processing disaster reports
- Managing rescue requests
- Handling notifications
- Database communication
- Business logic implementation
- Admin panel management

🔹 Database Technology

The project uses PostgreSQL as the primary database.

Database Benefits:

- Reliable data storage
- Secure transactions
- Efficient queries
- Structured relational schema
- Scalable for large datasets

🔹 Deployment Technology

The project can be deployed using:

- Vercel for frontend hosting
- Render for backend hosting

This ensures global accessibility, scalability, and continuous availability.

🛠️ Tools / Software Used

Various tools were used throughout the project lifecycle for development, testing, collaboration, and deployment.

Tool / Software	Purpose
Visual Studio Code	Code editor
Git	Version control
GitHub	Source code hosting
Postman	API testing
pgAdmin	Database management
Chrome Developer Tools	Frontend debugging
npm	Angular package management
pip	Python package management
Vercel	Frontend deployment
Render	Backend deployment

📊 Data Collection Approach

The Disaster Management System primarily uses **user-generated and administrator-managed data**.

🔹 Citizen Data Collection

Collected during registration:

- Full Name
- Email Address
- Mobile Number
- Password

🔹 Disaster Incident Data

Collected when users report emergencies:

- Type of disaster (Flood, Fire, Earthquake, etc.)
- Location
- Description
- Severity level
- Date and time
- Contact details

🔹 Rescue and Resource Data

Managed by administrators:

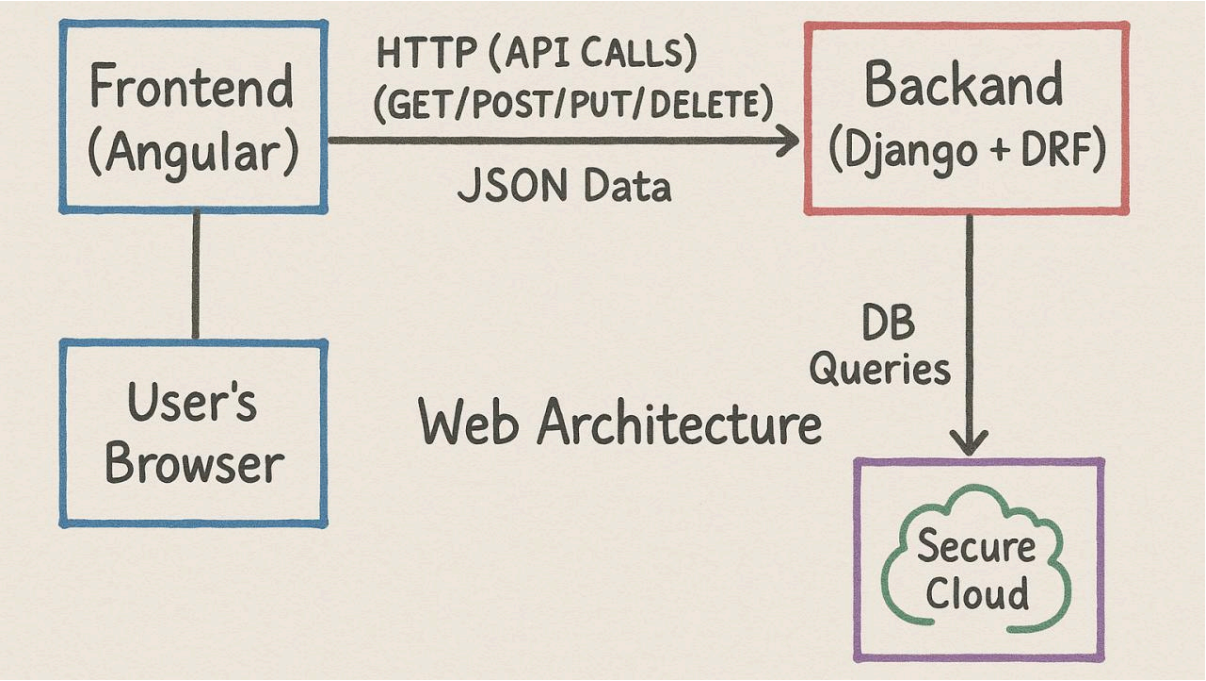
- Rescue team availability
- Shelter occupancy
- Ambulances and vehicles
- Food and medical resources

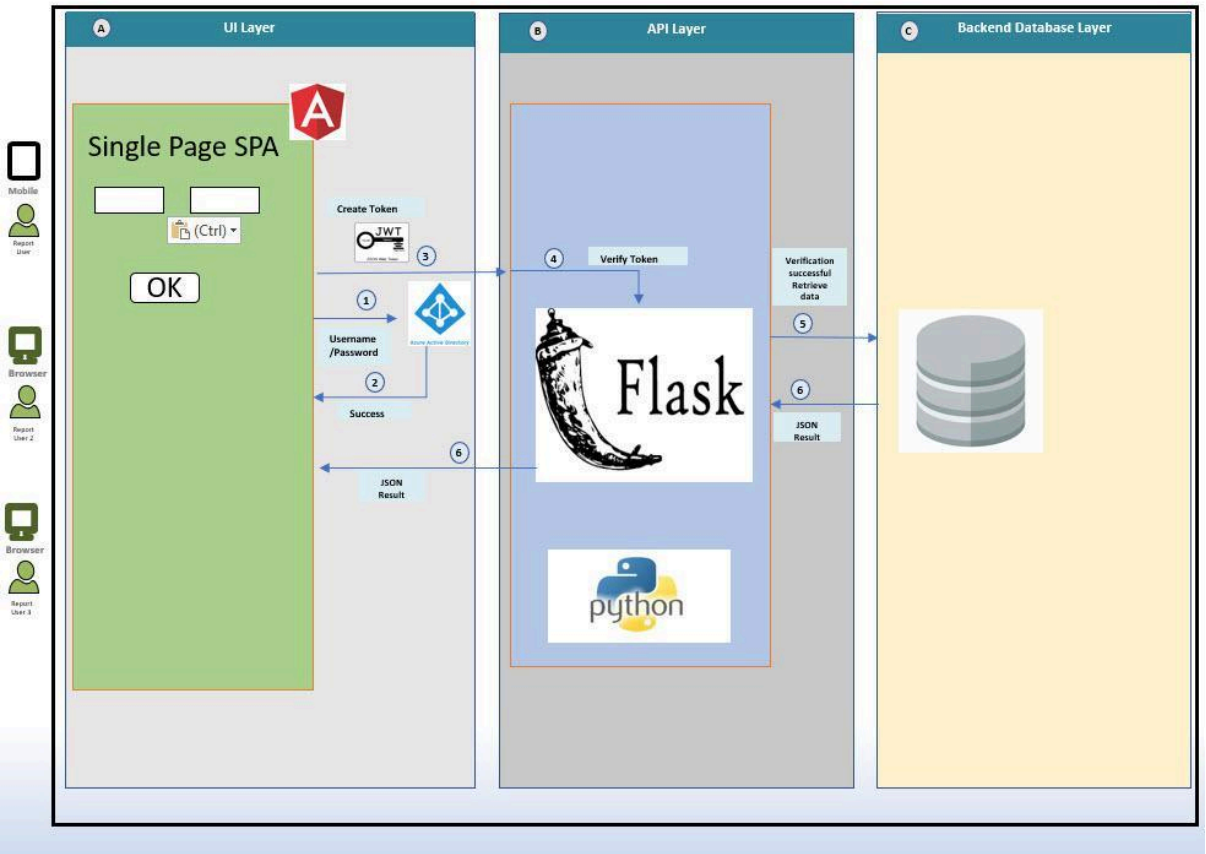
🔹 Future Data Enhancements

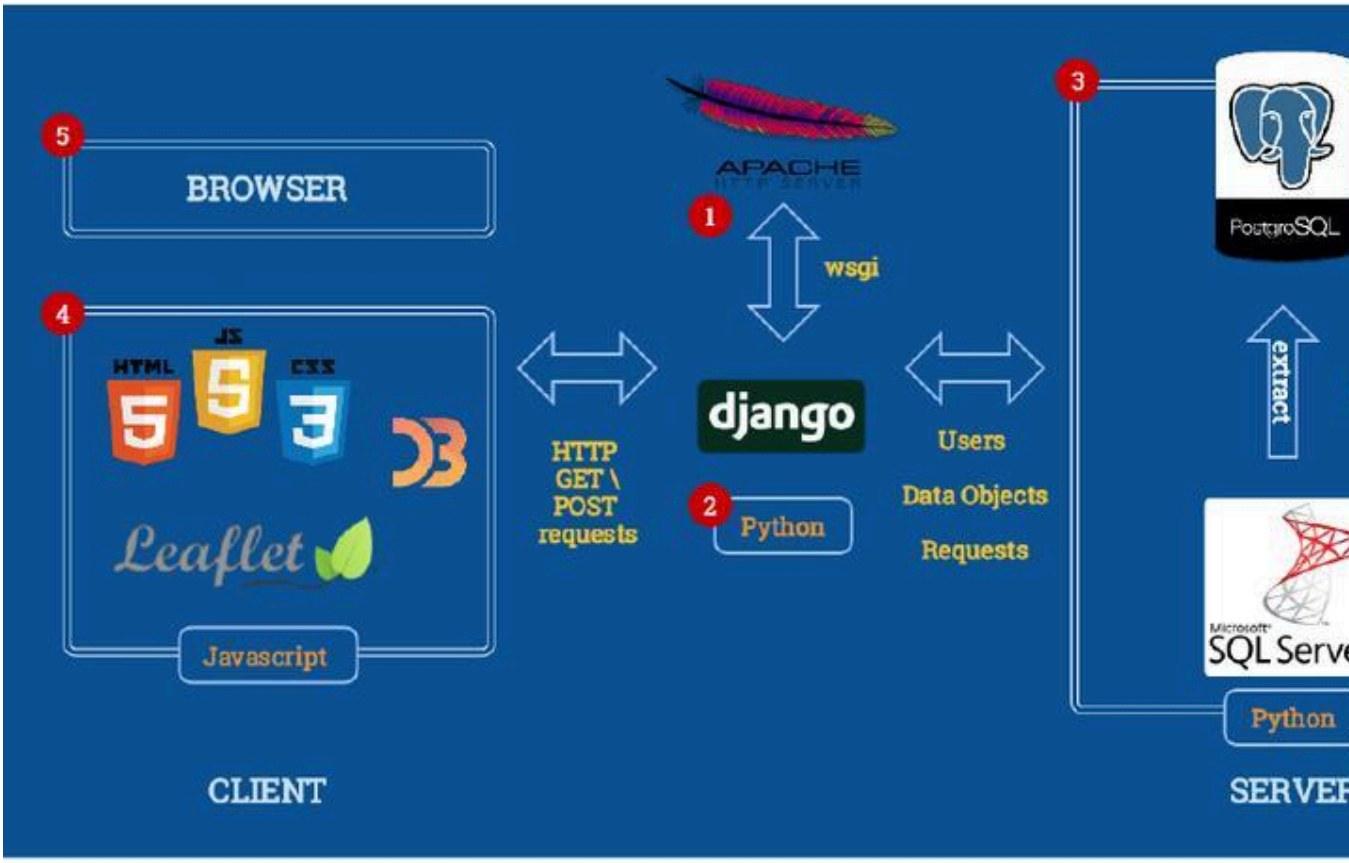
The system can later support:

- GPS location data
- Uploaded images/videos
- IoT sensor feeds
- Weather API data

🏗️ Project Architecture







The project follows a **Three-Tier Architecture**, which separates presentation, logic, and data layers for better maintainability and scalability.

🔹 Presentation Layer (Frontend)

Developed using Angular.

Responsibilities:

- User registration/login pages
- Disaster reporting forms
- Shelter information pages
- Rescue request pages
- Dashboard displays
- Alerts and notifications

Users interact directly with this layer.

🔹 Application Layer (Backend)

Developed using Django REST APIs.

Responsibilities:

- Process user requests
- Validate form data
- Authenticate users
- Manage disaster reports
- Allocate rescue resources
- Generate reports

This layer acts as the bridge between frontend and database.

🔹 Data Layer (Database)

Developed using PostgreSQL.

Stores:

- User accounts
- Disaster incidents
- Shelter records
- Rescue requests
- Notifications
- Admin logs

🔹 Workflow of the System

1. User logs into the system
2. Reports a disaster incident
3. Angular sends request to Django API
4. Django validates and stores data in PostgreSQL
5. Admin dashboard receives new alert
6. Rescue team assigned
7. Status updated in real time
8. User receives updates

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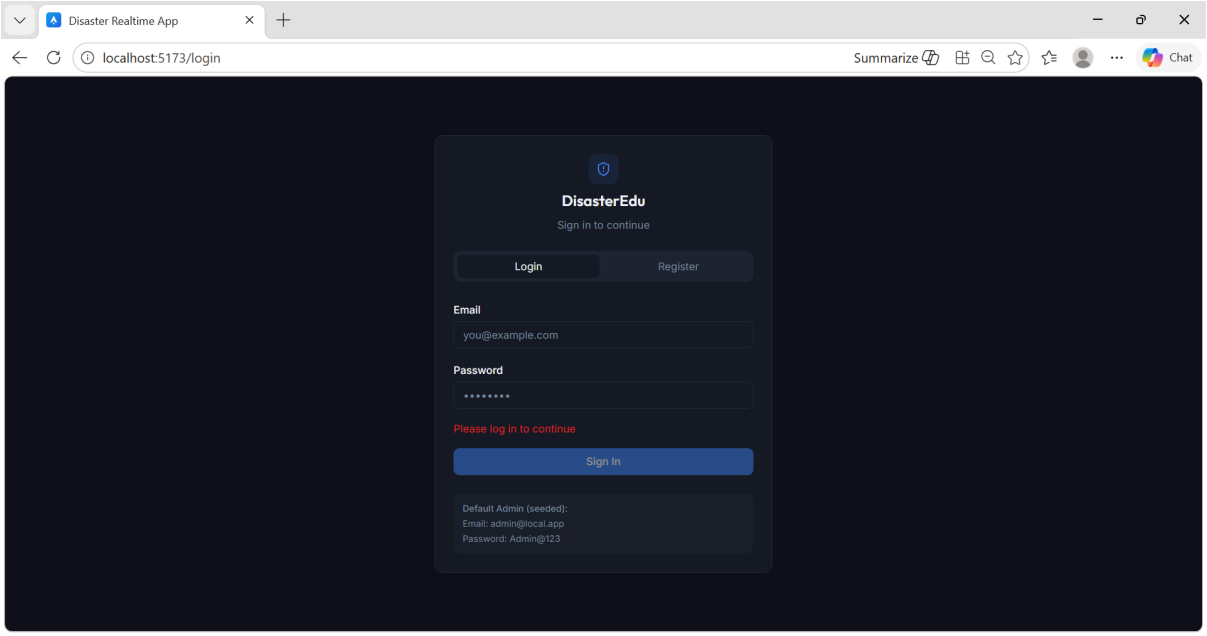
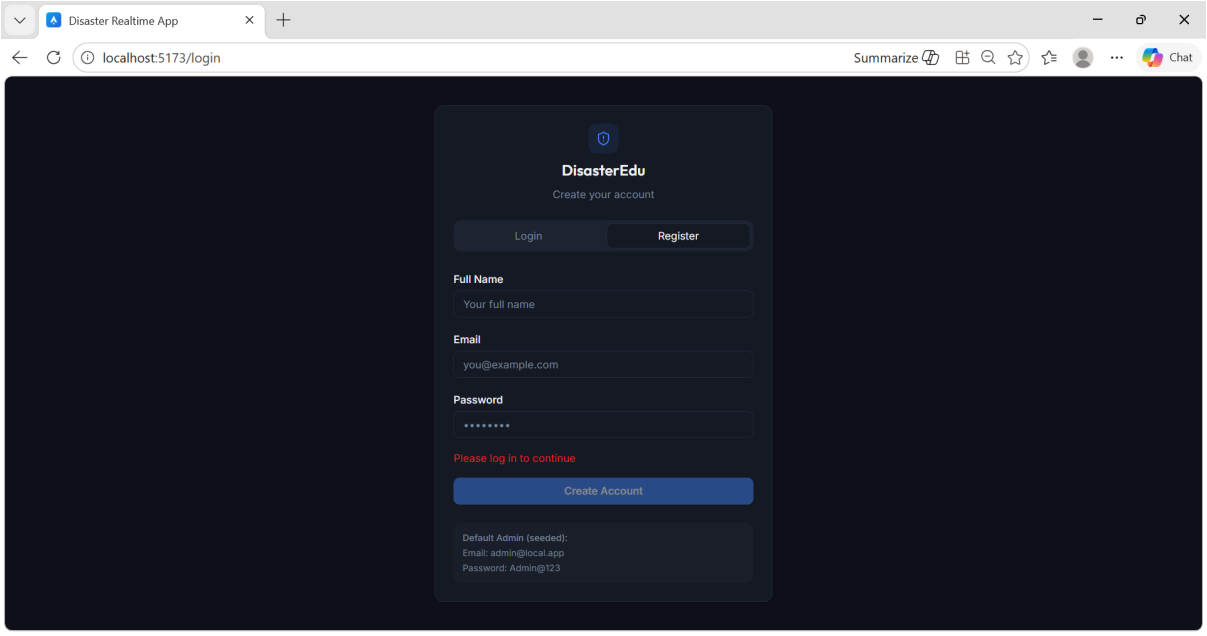
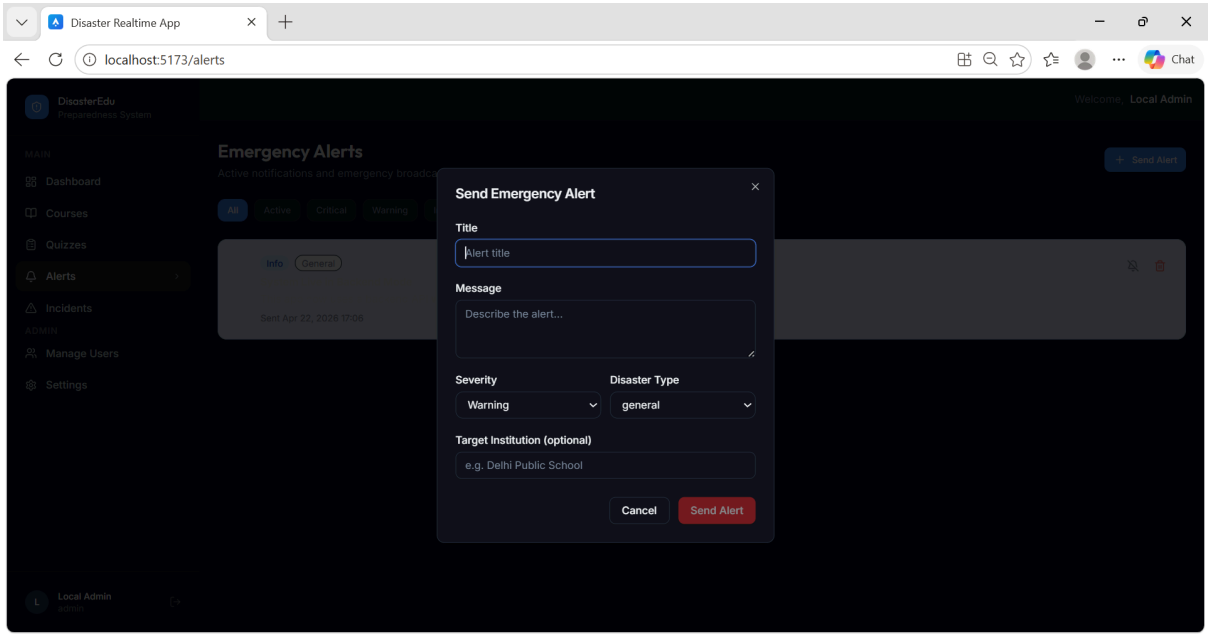
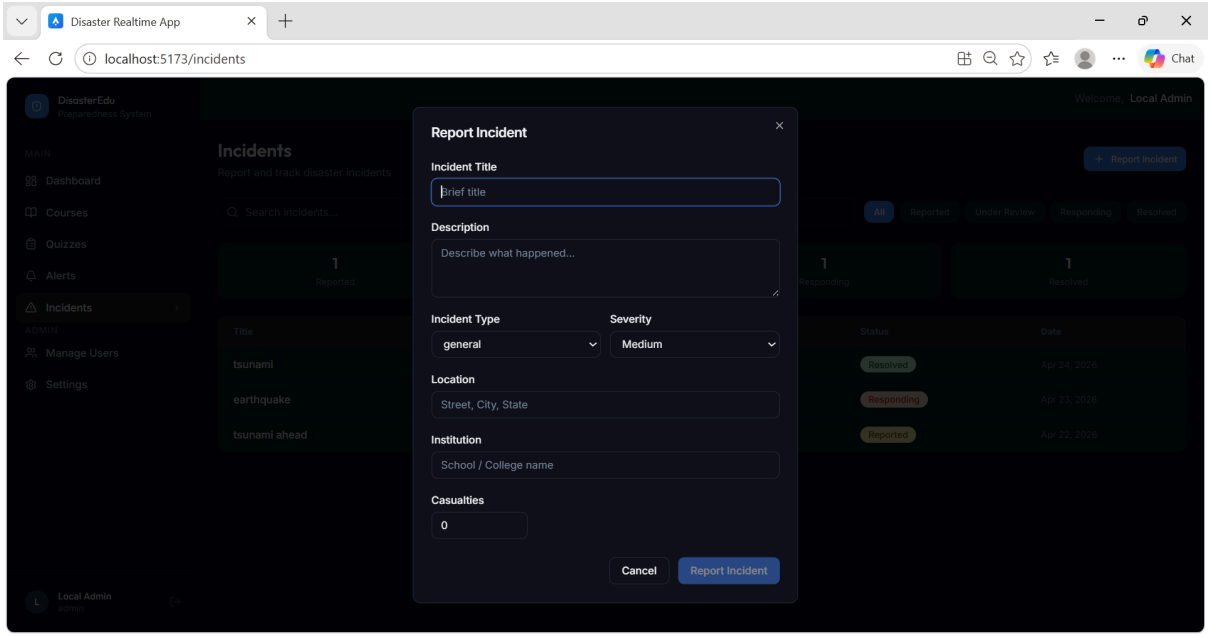
📸Final Project Working (Screenshots + Explanation)



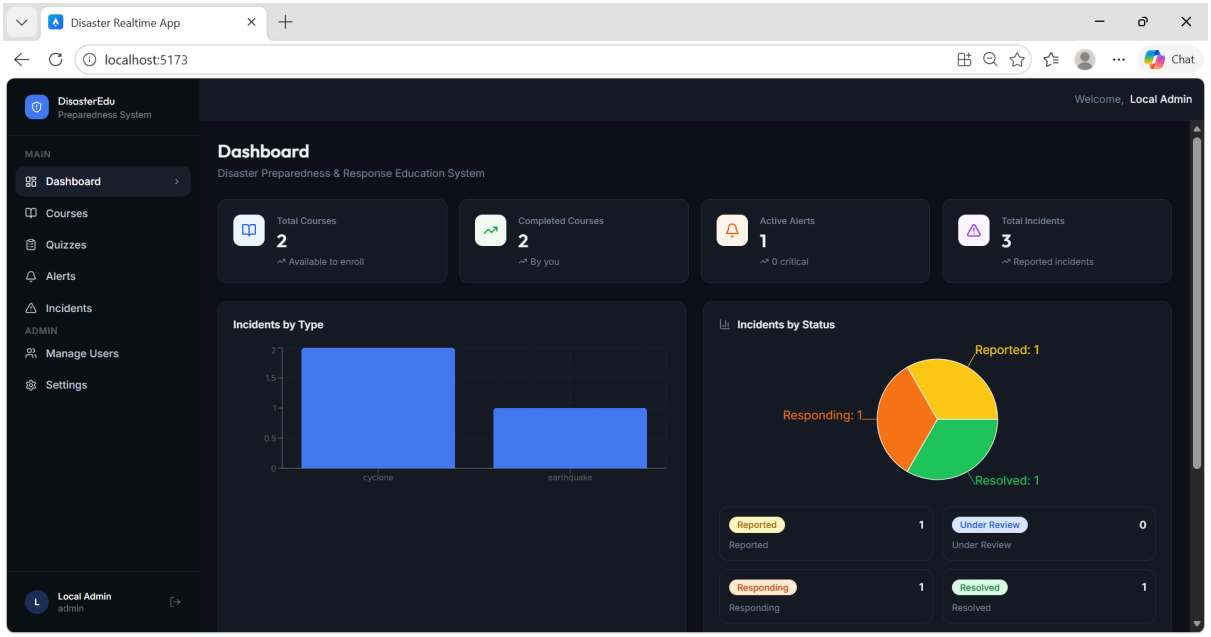
->log in page for authentication and to use essential features of the application

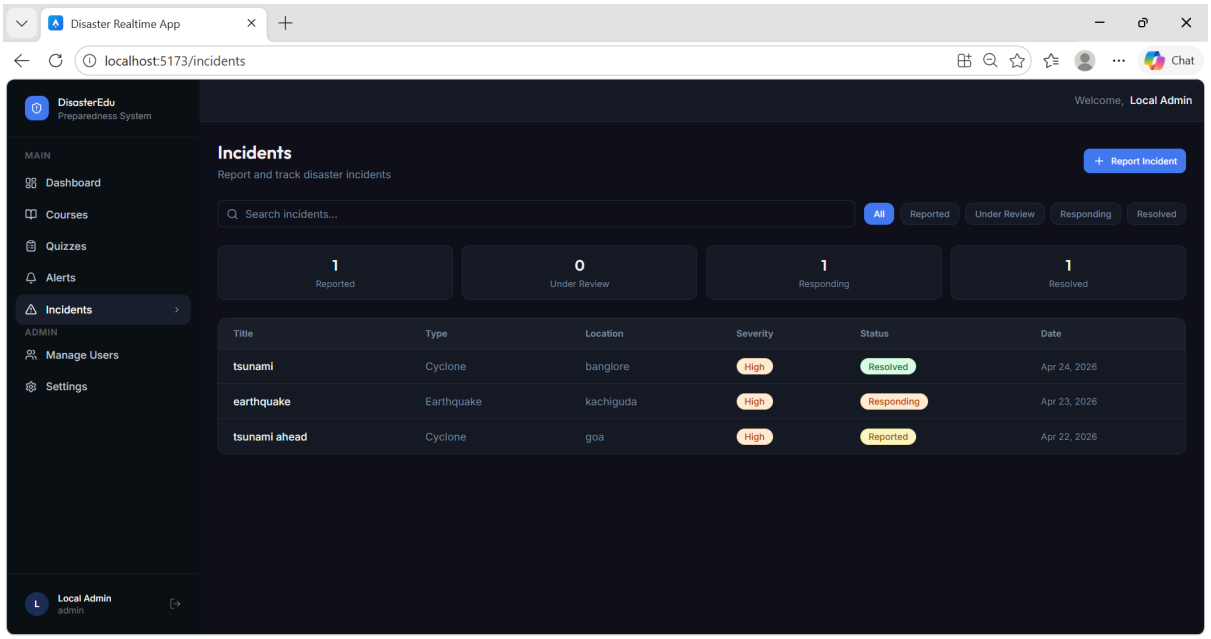




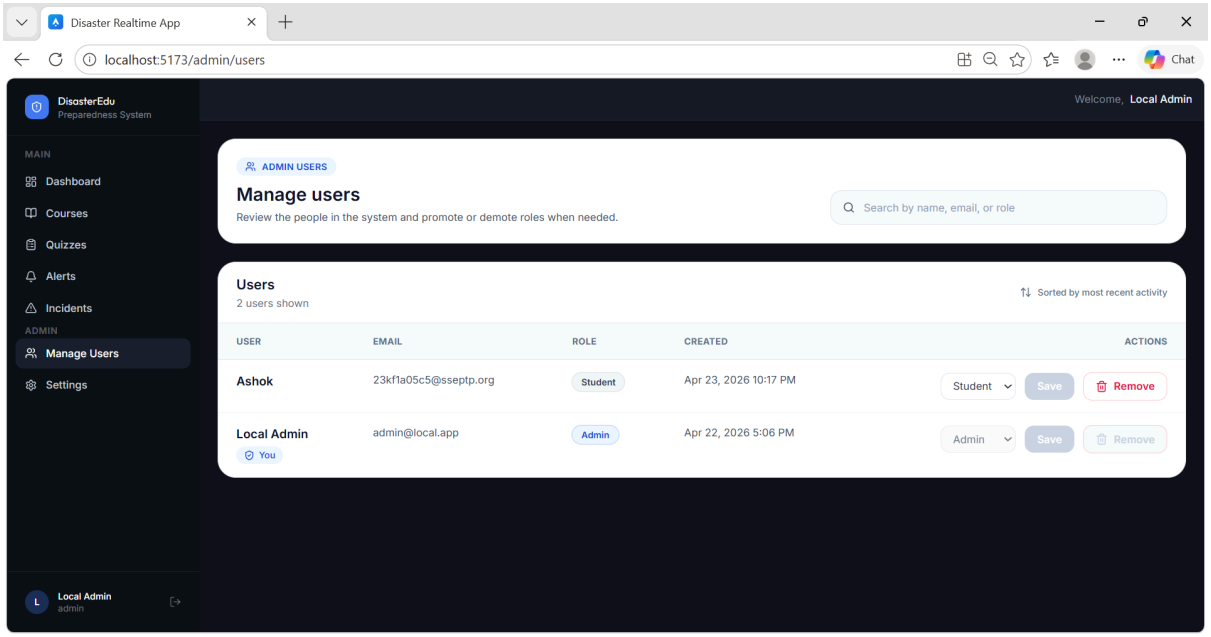


->Issue reporting is easier along with location and casualties affected.And issues are classified based on the severity.

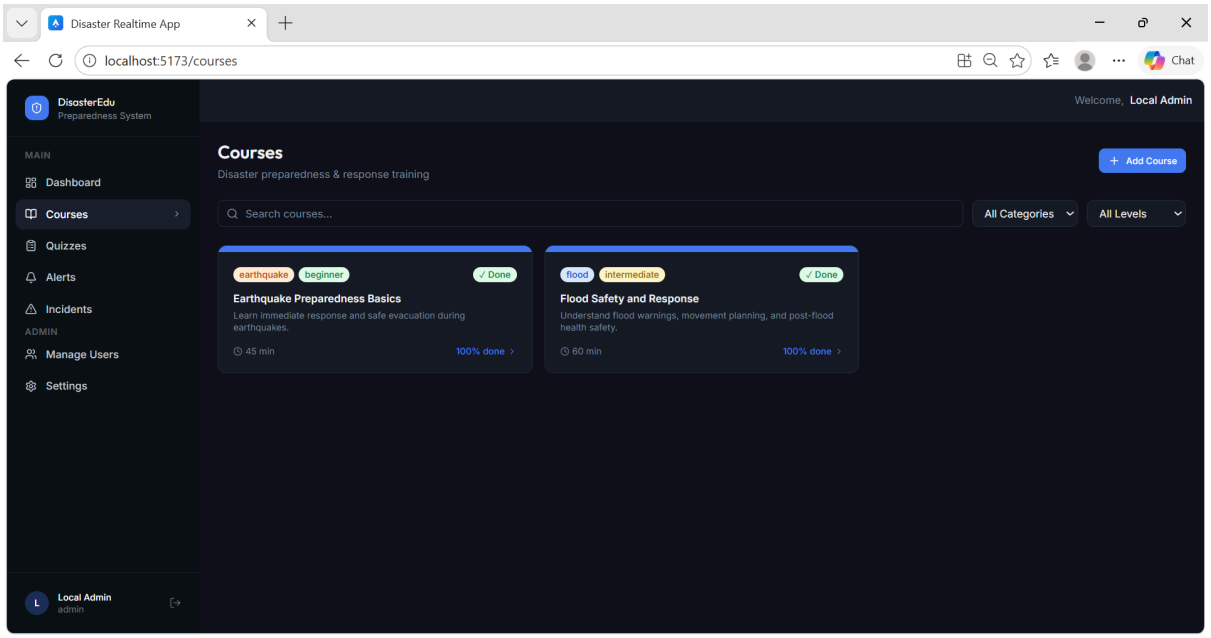




->A simple and clean dashboard for easier understanding of the issues.And to take preventive measures as soon as possible.



->Super admin has authority to make users as admins or remove them permanently.And also has access for setting the status of the issues.



->To educate users about the disaster and to make decisions to protect themselves at any such crisis.

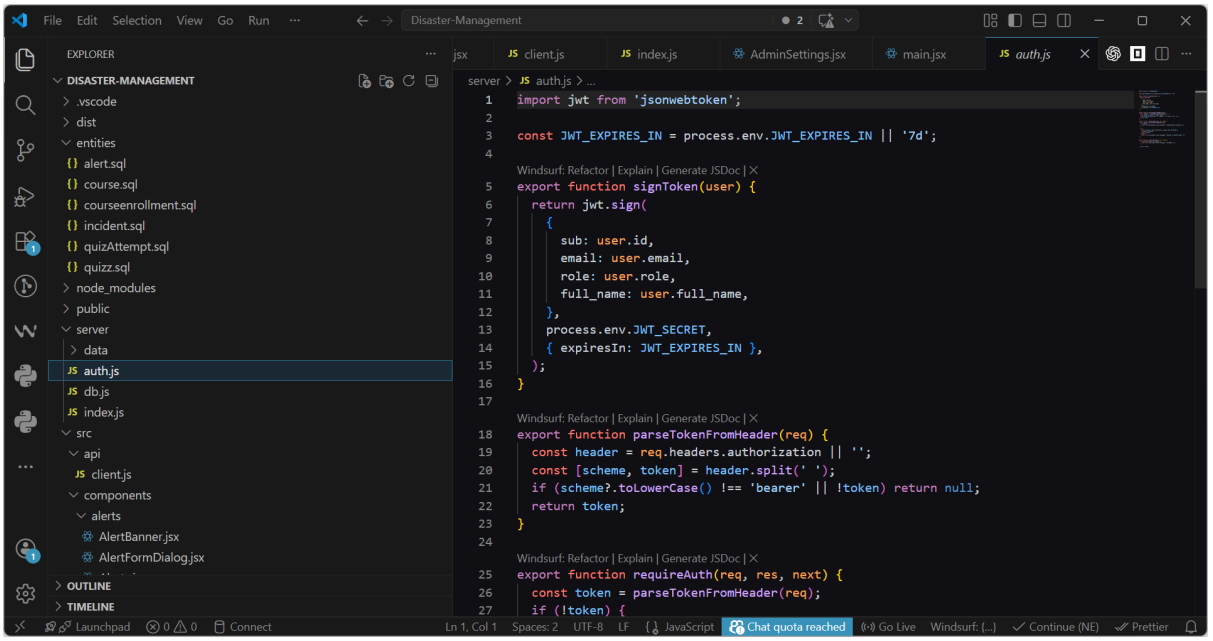
🔹 Key Screens:

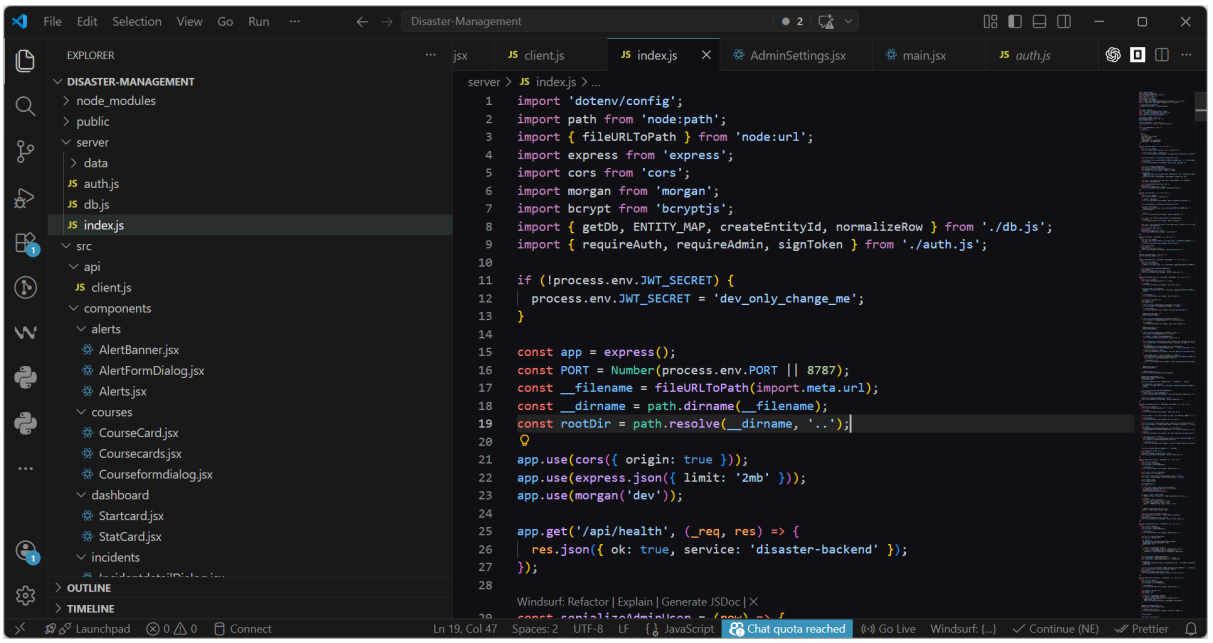
- **Login/Register Page:**
  Allows users to securely access the system
- **Issue Submission Form:**
  Users can report issues with details

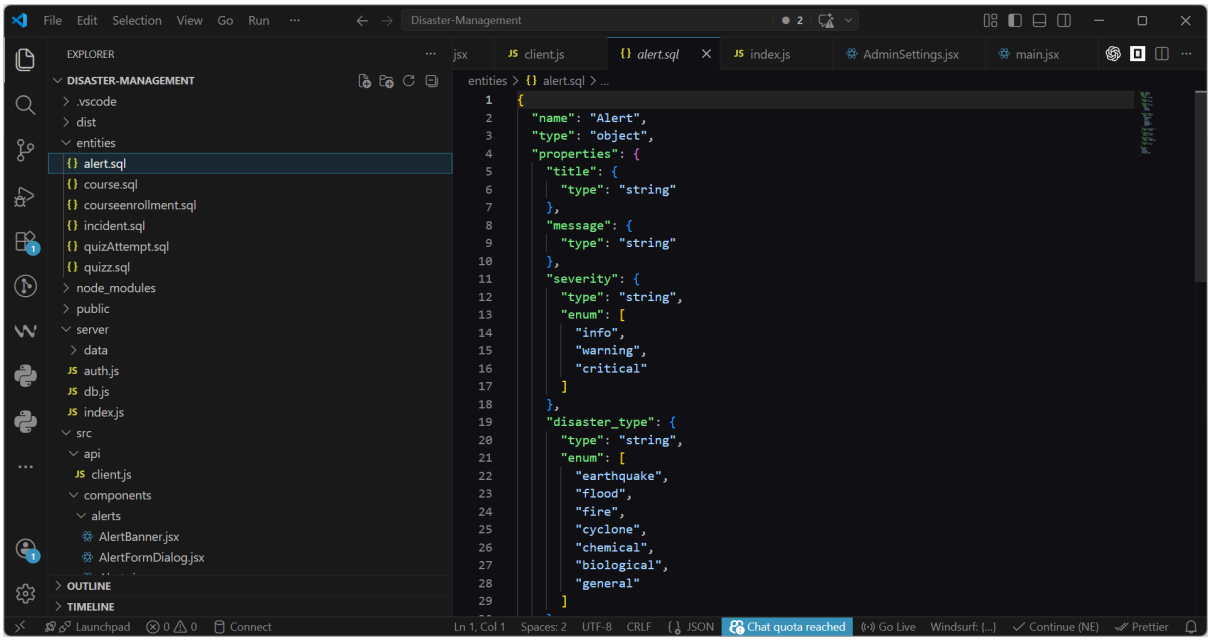
- **Dashboard:**
  Displays submitted issues and their status
- **Admin Panel:**
  Enables authorities to manage and update complaints

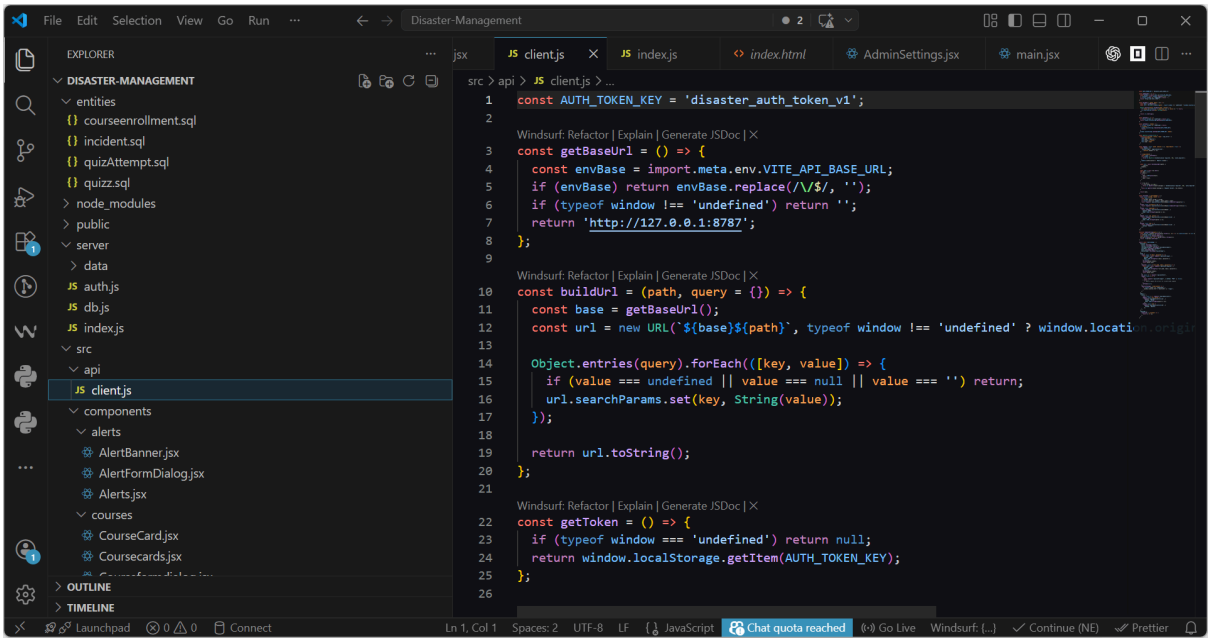
Each module is designed for simplicity and efficiency, ensuring a smooth user experience.

Code screenshots

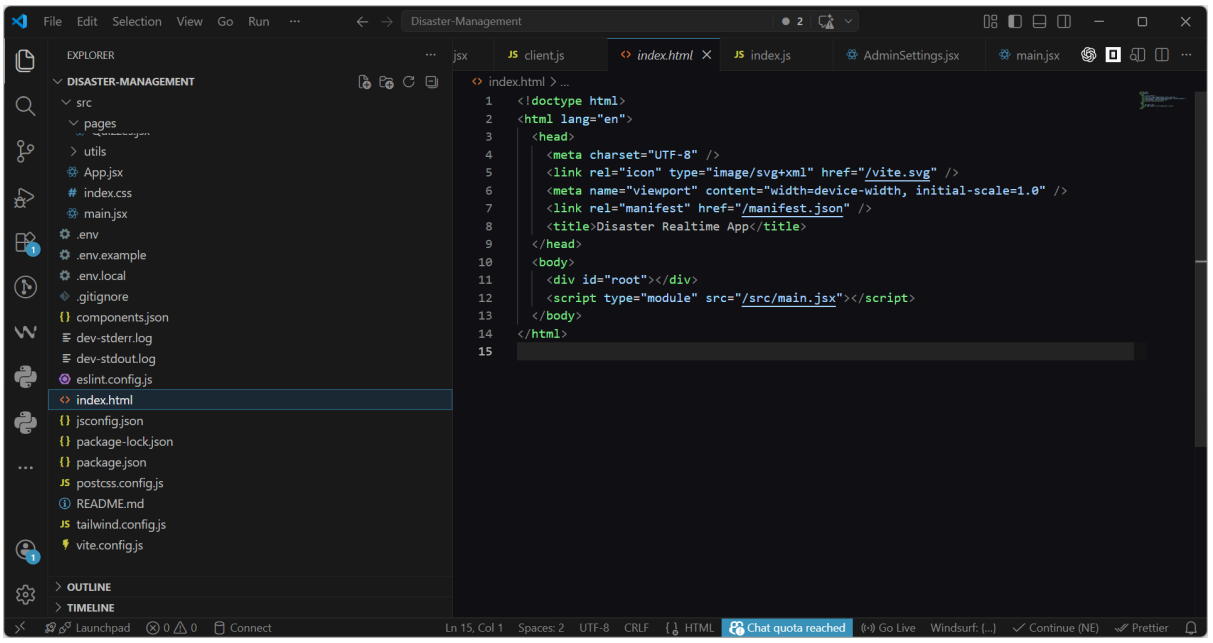








->Main code logic and api routing along with hyperlinks and database setup.



📈 Results

The project was successfully implemented and tested, achieving the following results:

- Functional web application with frontend and backend integration
- Real-time issue tracking and status updates
- Efficient data storage and retrieval
- Smooth deployment and accessibility via web

The system demonstrates the practical application of full-stack technologies in solving real-world problems effectively.

Social/Industry relevance of the project

The **Disaster Management System** has significant social and industrial relevance because it addresses one of the most critical needs of society: protecting human life and ensuring quick response during emergencies. In the modern world, where natural calamities and unexpected disasters are increasing due to climate change, urbanization, and industrial expansion, the need for technology-driven disaster preparedness and response systems has become more important than ever.

This project provides a digital platform that improves communication, coordination, and resource management during emergencies. It benefits citizens, local authorities, government departments, NGOs, healthcare institutions, and rescue agencies.

🔹 Helps Improve Urban Management

Modern cities face multiple risks such as floods, fires, building collapses, traffic accidents, industrial hazards, and waterlogging. Urban administrations require quick access to accurate information in order to respond effectively.

This system helps improve urban management by:

- Allowing instant reporting of emergencies
- Providing real-time dashboards for authorities
- Monitoring shelters and relief centers
- Coordinating municipal response teams
- Reducing delays in handling city emergencies
- Supporting better planning and disaster preparedness

By integrating digital tools into city administration, the project supports safer and more resilient urban environments.

🔹 Encourages Citizen Participation

Citizens play an important role during disasters. Early reporting of incidents, sharing local information, and requesting timely help can save lives.

This project encourages citizen participation by:

- Giving people a direct platform to report disasters
- Allowing users to request rescue assistance online

- Enabling citizens to track response status
- Providing alerts and public safety updates
- Building trust through transparent communication

When citizens actively participate, disaster management becomes faster, more accurate, and more effective.

🔹 Useful for Municipalities and Governments

Government bodies and municipal corporations are responsible for public safety, emergency services, and relief operations. The system can be highly useful for these organizations.

Applications include:

- Managing disaster control rooms
- Monitoring rescue requests
- Tracking emergency vehicles and teams
- Maintaining shelter databases
- Coordinating between departments
- Generating reports for decision-making

The project supports **digital governance** by replacing manual emergency handling methods with an automated system.

🔹 Can Be Used in Disaster Management Scenarios

The system is directly relevant for multiple disaster scenarios such as:

- Floods
- Cyclones
- Earthquakes
- Fires
- Landslides
- Building collapses
- Industrial accidents
- Pandemic emergencies

In each case, the platform can be used for:

- Incident reporting
- Rescue request handling
- Resource allocation

- Shelter management
- Public announcements
- Progress monitoring

This makes the system highly practical and adaptable.

🔹 Supports Smart City Initiatives

Smart cities rely on technology to improve governance, safety, transportation, and infrastructure management. Disaster readiness is one of the most important components of a smart city.

This project supports smart city initiatives by:

- Using digital dashboards and analytics
- Improving communication channels
- Providing real-time emergency data
- Enabling faster public service delivery
- Supporting future IoT and AI integrations

It contributes toward building intelligent, safe, and sustainable cities.

Learning and Reflection

The development of the Disaster Management System provided valuable technical, practical, and personal learning experiences. It helped in understanding how software solutions can solve real-world public safety challenges.

🔹 Learned Full-Stack Development

The project provided complete exposure to frontend, backend, and database technologies.

Frontend Learning:

Using Angular helped in learning:

- Component-based architecture
- Routing and navigation
- Form validation
- Responsive UI design
- Dashboard creation

Backend Learning:

Using Django helped in learning:

- REST API development
- Authentication systems
- Role management
- Data validation
- Business logic implementation

Database Learning:

Using PostgreSQL helped in understanding:

- Database schema design
- CRUD operations
- Relationships and queries
- Data security

🔹 Understood API Integration

One of the most valuable learnings was integrating frontend and backend systems using APIs.

This included:

- Sending HTTP requests
- Receiving JSON responses
- Displaying dynamic data
- Error handling
- Connecting multiple modules seamlessly

This experience improved understanding of real-world web application communication.

🔹 Gained Experience in Deployment

The project provided practical experience in deploying applications to cloud platforms such as:

- Vercel
- Render

Deployment helped in learning:

- Environment variables
- Build processes
- Production settings
- Hosting configuration
- Domain accessibility

🔹 Improved Debugging and Problem-Solving Skills

During development, several technical challenges were encountered such as:

- API connection errors
- CORS issues
- Authentication bugs
- Database migration problems
- Deployment failures
- UI responsiveness issues

Solving these problems improved:

- Logical thinking
- Troubleshooting ability
- Patience and persistence
- Independent learning skills

🔹 Personal Reflection

This project demonstrated that software engineering can create impactful solutions for society. It increased confidence in handling full-scale projects and highlighted the importance of teamwork, planning, and continuous improvement.

The **Disaster Management System** was developed with the primary objective of creating a modern, efficient, and reliable digital platform to improve disaster preparedness, emergency response, and coordination among citizens, rescue teams, and government authorities. In today's world, where natural and man-made disasters occur frequently, the importance of quick communication and organized management cannot be ignored. This project addresses these challenges by replacing traditional manual systems with a smart web-based solution.

The project successfully demonstrates how technology can be used to reduce disaster response time, improve transparency, and ensure better utilization of available resources. By integrating a user-friendly frontend, secure backend, and structured database system, the platform provides an effective way to report incidents, request assistance, track rescue operations, and manage emergency resources.

🔹 Recap of Objectives and Achievements

The project was designed with several important objectives, and these objectives were successfully achieved during development.

1. Creation of a Centralized Disaster Management Platform

One of the major goals was to create a single platform for managing all disaster-related operations. This objective was achieved by developing a unified system where users can report incidents, request help, and administrators can monitor activities through a dashboard.

2. Real-Time Incident Reporting

The system enables citizens to report disasters such as floods, fires, earthquakes, cyclones, accidents, and emergencies instantly through an online portal. This improves the speed at which authorities receive critical information.

3. Improved Communication and Coordination

The project successfully improves communication between affected citizens, rescue teams, and administrators. Centralized data and live updates help avoid confusion during emergency situations.

4. Efficient Resource Management

The platform can be used to manage:

- Rescue teams
- Ambulances
- Relief materials
- Shelter centers
- Medical facilities

This ensures better resource allocation during critical times.

5. Secure and Scalable Technology Implementation

The system was developed using modern technologies such as:

- Angular for frontend development
- Django for backend APIs
- PostgreSQL for database management

These technologies provide security, flexibility, and scalability.

6. Practical Full-Stack Web Application

The successful completion of this project proves the practical use of full-stack development for solving real-world problems in public safety and emergency management.

🔹 Major Achievements of the Project

- Functional disaster reporting system developed
- Real-time status tracking implemented
- Dashboard for authorities created
- Structured database system completed
- Frontend and backend successfully integrated
- Cloud deployment ready application developed
- Improved model for emergency communication demonstrated

🔹 Future Scope of this Project

Although the current system is effective, several advanced features can be added in future versions to increase efficiency and real-world usability.

📱 1. Mobile Application Development

A dedicated mobile application for Android and iOS can make emergency reporting easier and faster.

Benefits:

- Quick access during emergencies
- Push notifications
- Better accessibility in remote areas

🤖 2. AI-Based Disaster Prediction

Artificial Intelligence can be integrated for predicting disasters using historical and live data.

Examples:

- Flood prediction
- Fire risk alerts
- Cyclone severity analysis
- Risk zone identification

🗺️ 3. GPS and Map Integration

Map services can improve rescue operations through location tracking.

Features:

- Exact incident location mapping
- Rescue route optimization
- Nearby shelter suggestions
- Live movement tracking of rescue teams

🔔 4. Notification System (SMS / Email / Push Alerts)

Advanced alert systems can notify users instantly regarding:

- Evacuation orders
- Weather warnings
- Rescue status updates
- Shelter availability

📊 5. Advanced Analytics Dashboard

Government authorities can use analytics dashboards to monitor trends.

Includes:

- Disaster frequency reports
- Response time statistics
- Resource utilization charts
- Region-wise risk analysis

🌐 6. Multi-Language Support

To improve accessibility, the system can support multiple regional languages so more citizens can use it comfortably.

🚁 7. Drone and IoT Integration

Future versions may integrate:

- Drone surveillance during floods/fires
- IoT flood level sensors
- Fire detection alarms
- Smart warning systems

Final Conclusion

The Disaster Management System is a valuable and practical project that modernizes traditional emergency response systems. It improves disaster readiness, communication, coordination, and public safety through digital technology.

The successful implementation of this project highlights the power of Angular, Django, and PostgreSQL in creating scalable and impactful real-world solutions.

With future enhancements such as mobile apps, AI prediction, GPS rescue tracking, and analytics, this project has the potential to become a comprehensive large-scale disaster management platform capable of serving cities, states, and even national emergency agencies.